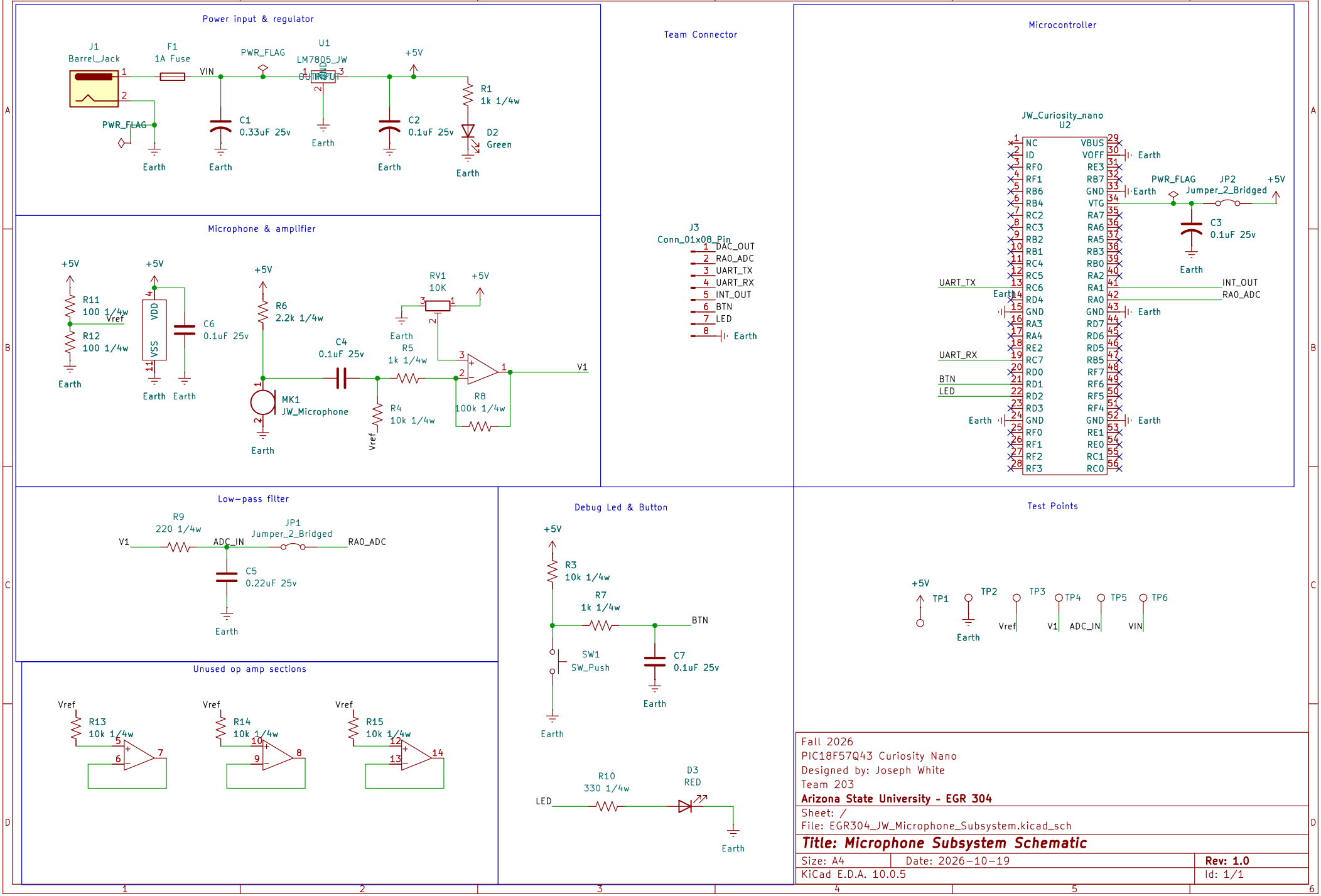

Power input & regulator
J1
Barrel_Jack
F1
1A Fuse
VIN
PWR_FLAG
U1
LM7805_JW
IN
OUT
+5V
R1
1k 1/4w
D2
Green
C1
0.33uF 25v
C2
0.1uF 25v
Earth
Team Connector
Microcontroller
JW_Curiosity_nano
U2
NC
ID
RF0
RF1
RB6
RB4
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0
RF1
RF2
RF3
VBUS
VOFF
RE3
RB7
GND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0
PWR_FLAG
JP2
Jumper_2_Bridged
C3
0.1uF 25v
UART_TX
UART_RX
BTN
LED
INT_OUT
RA0_ADC
Microphone & amplifier
R11
100 1/4w
R12
100 1/4w
Vref
VDD
VSS
C6
0.1uF 25v
R6
2.2k 1/4w
C4
0.1uF 25v
RV1
10K
R5
1k 1/4w
MK1
JW_Microphone
R4
10k 1/4w
R8
100k 1/4w
V1
J3
Conn_01x08_Pin
DAC_OUT
RA0_ADC
UART_TX
UART_RX
INT_OUT
BTN
LED
Earth
Low-pass filter
R9
220 1/4w
JP1
Jumper_2_Bridged
ADC_IN
C5
0.22uF 25v
Debug Led & Button
R3
10k 1/4w
R7
1k 1/4w
SW1
SW_Push
C7
0.1uF 25v
R10
330 1/4w
D3
RED
Test Points
TP1
TP2
TP3
TP4
TP5
TP6
Unused op amp sections
R13
10k 1/4w
R14
10k 1/4w
R15
10k 1/4w
Fall 2026
PIC18F57Q43 Curiosity Nano
Designed by: Joseph White
Team 203
Arizona State University - EGR 304
Sheet: /
File: EGR304_JW_Microphone_Subsystem.kicad_sch
Title: Microphone Subsystem Schematic
Size: A4
Date: 2026-10-19
Rev: 1.0
KiCad E.D.A. 10.0.5
Id: 1/1